

Model Recommendation Memo

Objectives → The objectives was to predict Titanic passenger survival rate using six available features **i.e. pclass, sex_male, age, sibsp, parch, fare**. The dataset contains 1000 rows and stratified it into 80/20 train/test split with random_state=42. The test set retained as the final external evaluation set.

Model Choice → Random Forest is a suitable baseline because it combines many decision trees trained with bootstrap sampling and random feature subsets. This reduces the variance of individual decision trees while retaining the ability to model nonlinear relationships and feature interactions. It also provides both class prediction and survival probabilities.

Evaluation → The final random forest used 200 trees and "sqrt", "log2", 0.5, and 1.0. for max_features. Its final test was 0.73 and ROC-AUC 0.8315 and OOB Score was 0.7087 with a standard deviation of 0.8639, indicating good stability.

Key Findings → Permutation importance showed that Sex_male and fare were among the most useful features for held-out predictive performance. The PDP for Sex_male showed how many male survived it gives 1 to male and 0 for female, while the PDP for fare showed the classes. These findings describe relationships learned by the model and should not be interpreted as casual effects.

Risks and Limitations → The Titanic dataset is historical and relatively small, so its pattern may not be generalize to other populations or situations. Missing ages required group-based median imputation, which removes some individual-level variation. Feature importance also has methodological limitations, particularly for impurity-based importance. In addition, correlation does not establish causation, and PDPs may average over unrealistic combinations of correlated features. Finally, test-set performance contains sampling uncertainty and should not be treated as proof of universal model superiority.

Recommendation → I recommend to use Random forest as a useful baseline, but not as the final model without further validation. The next step should be cross-validation on the training data, followed by comparison with logistic regression and a boosting model. Probability calibration could also be investigated if the predicted survival probabilities will be used for decision making.